



## Solutions:

### 1. A. Zeroth Law of Thermodynamics

#### Solution:

That is the most common statement used to denote the Zeroth Law of Thermodynamics or the Law of Thermal Equilibrium.

Reference: GEAS 03-02 – The Zeroth Law of Thermodynamics

### 2. B. First Law of Thermodynamics

#### Solution:

The First Law of Thermodynamics is also an extension of the Law of Conservation of Energy. If the statement implies that the energy is conserved, that is automatically part of the First Law of Thermodynamics.

Reference: GEAS 03-03- Internal Energy, Heat and Work

### 3. A. 240 J absorbed

#### Solution:

$$\begin{aligned} \Delta U &= \pm Q \pm W \\ +560 &= Q + 320 \\ Q &= +240\text{J} \rightarrow \text{Ans} \end{aligned}$$

Sign Convention:

- + Q : heat enters the system
- Q : heat leaves the system
- + W : work done on the system
- W : work done by the system

Also, if the internal energy increased, the change in internal energy should be positive.

Reference: GEAS 03-03- Internal Energy, Heat and Work

### 4. C. Second Law of Thermodynamics

#### Solution:

According to the 2<sup>nd</sup> Law of Thermodynamics, the entropy of any real process is always increasing.

Reference: GEAS 03-12 – Entropy and the 2<sup>nd</sup> Law of Thermodynamics

### 5. C. 25%

#### Solution:

$$\begin{aligned} e &= 1 - \frac{Q_c}{Q_h} \\ e &= 1 - \frac{600}{800} = 25\% \rightarrow \text{Ans} \\ W_{\text{eng}} &= Q_h - Q_c \\ W_{\text{eng}} &= 800 - 600 = 200\text{J} \end{aligned}$$

Reference: GEAS 03-09 – Heat Engines

### 6. B. rms speed increases by a factor of 2

#### Solution:

According to the formula or rms speed, the speed is directly proportional to the square root of the absolute temperature.

$$\begin{aligned} T_1 &= -73^\circ\text{C} + 273 = 200\text{K} \\ T_2 &= 527^\circ\text{C} + 273 = 800\text{K} \end{aligned}$$

If the absolute temperature increased by a factor of 4, the rms speed will increase by a factor of  $\sqrt{4}$  or 2.

Reference: GEAS 03-06 – Kinetic Theory of Gases

### 7. B. P will increase by a factor of 4

#### Solution:

According to the Ideal Gas Law, pressure is directly proportional to absolute temperature. If the temperature increases by 4 times, the pressure also increases by a factor of 4.

Reference: GEAS 03-06 – Kinetic Theory of Gases

### 8. D. P will increase by a factor of 2

#### Solution:

According to the Ideal Gas Law (or Boyle's Law), the pressure is inversely related to volume when the temperature is constant. So if the volume is halved, the pressure should double.

Reference: GEAS 03-06 – Kinetic Theory of Gases

### 9. B. 180°C and 13°C

#### Solution:

For Gold:

$$\begin{aligned} Q &= mc\Delta T \\ 2340\text{J} &= (0.1\text{kg}) \left( 130 \frac{\text{J}}{\text{kg}^\circ\text{C}} \right) \Delta T \\ \Delta T &= 180^\circ\text{C} \rightarrow \text{Ans} \end{aligned}$$

For wood:

$$\begin{aligned} Q &= mc\Delta T \\ 2340\text{J} &= (0.1\text{kg}) \left( 1800 \frac{\text{J}}{\text{kg}^\circ\text{C}} \right) \Delta T \\ \Delta T &= 13^\circ\text{C} \rightarrow \text{Ans} \end{aligned}$$

Reference: GEAS 03-04 – Conservation of Energy

### 10. B. 33.4 kJ

#### Solution:

Latent heat of fusion is the energy needed to change phase from solid or liquid (and vice versa).

$$\begin{aligned} Q &= mL_f \\ Q &= (0.1\text{kg}) \left( 334 \frac{\text{kJ}}{\text{kg}} \right) = 33.4\text{kJ} \rightarrow \text{Ans} \end{aligned}$$

Make sure you memorize all the latent heat values of water. Usually, it's not given in the problem.

Reference: GEAS 03-05 – Latent and Sensible Heat

### 11. C. 30°C

#### Solution:

By conservation of energy, the amount of energy lost by the teflon is the same amount of energy gained by the water:

$$\begin{aligned} -Q_{\text{teflon}} &= +Q_{\text{water}} \\ -(mc\Delta T)_{\text{teflon}} &= +(mc\Delta T)_{\text{water}} \\ -(0.05)c_{\text{teflon}}(T_f - 430) &= +(0.5)4c_{\text{teflon}}(T_f - 20)_{\text{water}} \\ -(0.05)(T_f - 430) &= +(0.5)4(T_f - 20)_{\text{water}} \\ T_f &= 30^\circ\text{C} \rightarrow \text{Ans} \end{aligned}$$

Reference: GEAS 03-04 – Conservation of Energy



12. C. 264 kJ

**Solution:**

$$Q_{\text{total}} = Q_{\text{sensible}} + Q_{\text{latent}}$$

$$Q_{\text{total}} = mc\Delta T + mL_v$$

$$Q_{\text{total}} = (0.1\text{kg})\left(4186\frac{\text{J}}{\text{kg}^\circ\text{C}}\right)(100 - 20) + (0.1\text{kg})\left(2300\frac{\text{kJ}}{\text{kg}}\right)$$

$$Q_{\text{total}} = 264 \text{ kJ} \rightarrow \text{Ans}$$

Reference: GEAS 03-05 – Latent and Sensible Heat

13. B. 7.2 mm

**Solution:**

$$DL = aLDT$$

$$DL = (1.2 \times 10^{-5} / ^\circ\text{C})(30\text{m})(35 - 15)$$

$$DL = 7.2\text{mm} \rightarrow \text{Ans}$$

$$L_f = 30\text{m} + 7.2\text{mm} = 30.0072\text{m} \rightarrow \text{Ans}$$

Reference: GEAS 03-01 – Temperature and Thermal Expansion

14. D. 1.8 cm

**Solution:**

Solve first for the volume expansion of the mercury in the thermometer:

$$DV = bVDT$$

$$DV = (1.8 \times 10^{-4})(5 \times 10^{-7})(20)$$

$$DV = 1.8 \times 10^{-9}\text{m}^3$$

The increase in volume of the mercury in the thermometer is the reason why the mercury “rises”. The change in volume by expansion is also:

$$DV = 1.8 \times 10^{-9}\text{m}^3$$

$$Ah = 1.8 \times 10^{-9}\text{m}^3$$

$$(1 \times 10^{-7})h = 1.8 \times 10^{-9}\text{m}^3$$

$$h = 0.018\text{m} \rightarrow \text{Ans}$$

Reference: GEAS 03-01 – Temperature and Thermal Expansion

15. D. The entropy of the universe is always increasing.

**Solution:**

According to the 2<sup>nd</sup> Law of Thermodynamics, the entropy of any real process is always increasing.

Reference: GEAS 03-12 – Entropy and the 2<sup>nd</sup> Law of Thermodynamics

16. A. 1/4

**Solution:**

$$e = 1 - \frac{Q_c}{Q_h}$$

$$e = 1 - \frac{300}{400} = 25\% \rightarrow \text{Ans}$$

Reference: GEAS 03-09 – Heat Engines

17. A. 2

**Solution:**

According to the Ideal Gas Law, if the absolute temperature is doubled, the pressure must double as well.

Reference: GEAS 03-06 – Kinetic Theory of Gases

18. D. 2,400 cm<sup>3</sup>

**Solution:**

$$DV = bVDT$$

$$DV = (4 \times 10^{-5} / ^\circ\text{C})(2\text{m}^3)(35^\circ\text{C} - 5^\circ\text{C})$$

$$DV = 2.4 \times 10^{-3}\text{m}^3 \rightarrow \text{Ans}$$

Reference: GEAS 03-01 – Temperature and Thermal Expansion

19. A. 100 degrees Celsius

**Solution:**

Solve first for the amount of energy needed to raise the temperature from 20 to 100 degrees:

$$Q = mc\Delta T$$

$$Q = (0.1\text{kg})\left(4.186\frac{\text{kJ}}{\text{kg}^\circ\text{C}}\right)(80^\circ\text{C})$$

$$Q = 33.49\text{kJ}$$

From the 100 kJ of heat absorbed by the water, you are left with 66.51 kJ since the 33.49 kJ is already used up in increasing the temperature.

Solve for the amount of energy needed to transform all of the liquid into gas.

$$Q = mL_v$$

$$Q = (0.1\text{kg})\left(2260\frac{\text{kJ}}{\text{kg}}\right)$$

$$Q = 226\text{kJ}$$

You don't have enough energy for the liquid water to completely transform into gas. Therefore, with 100 kJ, the final temperature is 100 degrees Celsius.

Reference: GEAS 03-05 – Latent and Sensible Heat

20. A. 0.01 kg

**Solution:**

$$Q = mL_f$$

$$2000 = m\left(200\frac{\text{kJ}}{\text{kg}}\right)$$

$$m = 0.01\text{kg} \rightarrow \text{Ans}$$

Reference: GEAS 03-05 – Latent and Sensible Heat

21. A. 600 J

**Solution:**

$$Q = mc\Delta T$$

$$Q = (0.04\text{kg})\left(0.50\frac{\text{kJ}}{\text{kg}^\circ\text{C}}\right)(30^\circ\text{C})$$

$$Q = 0.6 \text{ kJ} \rightarrow \text{Ans}$$

Reference: GEAS 03-01 – Temperature and Thermal Expansion

22. A. 21.8°C

**Solution:**

By conservation of energy, the amount of energy lost by the gold nugget is the same amount of energy gained by the water:



$$-Q_{\text{gold}} = +Q_{\text{water}}$$

$$-(mc\Delta T)_{\text{gold}} = +(mc\Delta T)_{\text{water}}$$

$$-(0.05)(0.129)(T_f - 600) = +(0.5)(4.186)(T_f - 20)_{\text{water}}$$

$$T_f = 21.8^\circ\text{C} \rightarrow \text{Ans}$$

Reference: GEAS 03-04 –  
Conservation of Energy

23. C. 293 K

**Solution:**

Convert 68F to Celsius, then to Kelvin.

Reference: GEAS 03-01 –  
Temperature and Thermal Expansion

24. D. 96 K

**Solution:**

$$Q = nc_v \Delta T$$

$$1000\text{J} = (0.5\text{mol}) \left( 20.8 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \Delta T$$

$$\Delta T = 96.2\text{K}$$

Reference: GEAS 03-06 – Kinetic  
Theory of Gases

25. B. 69 K

**Solution:**

$$Q = nc_p \Delta T$$

$$1000\text{J} = (0.5\text{mol}) (c_v + R) \Delta T$$

$$1000\text{J} = (0.5\text{mol}) \left( \frac{5}{2} R + R \right) \Delta T$$

$$1000\text{J} = (0.5\text{mol}) \left( \frac{7}{2} R \right) \Delta T$$

$$1000\text{J} = (0.5\text{mol}) \left( \frac{7}{2} \left( 8.413 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \right) \Delta T$$

$$\Delta T = 68.73\text{K} \rightarrow \text{Ans}$$

Reference: GEAS 03-06 – Kinetic  
Theory of Gases

26. C. 350 J

**Solution:**

$$W_{\text{eng}} = Q_h - Q_c$$

$$W_{\text{eng}} = 800 - 450 = 350\text{J}$$

Reference: GEAS 03-09 – Heat  
Engines

27. D. they cannot be at different  
temperatures

**Solution:**

Thermal Equilibrium means the  
temperatures are the same.

Reference: GEAS 03-01 –  
Temperature and Thermal Expansion

28. B. 23.9 degrees Celsius

**Solution:**

The amount of energy lost by the  
brass is the same amount of energy  
gained by the water. Note that  
energy flows from the hotter body to  
the colder body. By conservation of  
energy:

$$-Q_{\text{hot}} = +Q_{\text{cold}}$$

$$-(mc\Delta T)_{\text{hot}} = +(mc\Delta T)_{\text{cold}}$$

$$-\left( (80\text{g}) \left( 0.092 \frac{\text{cal}}{\text{gK}} \right) (T - 292^\circ\text{C}) \right) = +\left( (200\text{g}) \left( 1 \frac{\text{cal}}{\text{gK}} \right) (T - 14^\circ\text{C}) \right)$$

$$T = 23.9^\circ\text{C} \rightarrow \text{Ans}$$

Note that in the equation,

$$\Delta T_K = \Delta T_C$$

Reference: GEAS 03-04 –  
Conservation of Energy

29. B. 54.8 psia

**Solution:**

Note that a pressure gauge measures  
gauge pressure, that is, the pressure  
in excess of atmospheric pressure (1  
atm = 14.7 psi). Thus, the initial  
pressure is 46.7 psi or 46.7 psia.

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$\frac{46.7}{P_2} = \frac{4 + 273}{50 + 273}$$

$$P_2 = 54.8\text{psia} \rightarrow \text{Ans}$$

Reference: GEAS 03-06 – Kinetic  
Theory of Gases

30. B. 150 J

**Solution:**

$$e = 1 - \frac{T_c}{T_h} = 1 - \frac{270}{360} = 0.25$$

$$e = \frac{W_{\text{eng}}}{Q_h}; 0.25 = \frac{W_{\text{eng}}}{600}$$

$$W_{\text{eng}} = 150\text{J} \rightarrow \text{Ans}$$

Reference: GEAS 03-09 – Heat  
Engines

31. D. 1.3 kJ/K

**Solution:**

$$\Delta S = \int \frac{dQ}{T} = \int_{273}^{373} \frac{mc dT}{T} = mc \ln \left( \frac{373}{273} \right)$$

$$\Delta S = (1\text{kg}) \left( 4.186 \frac{\text{kJ}}{\text{kgK}} \right) \ln \left( \frac{373}{273} \right)$$

$$\Delta S = 1.3 \frac{\text{kJ}}{\text{K}} \rightarrow \text{Ans}$$

Reference: GEAS 03-12 – Entropy  
and the 2<sup>nd</sup> Law of Thermodynamics

32. C.  $6.4 \times 10^{17}\text{W}$

**Solution:**

Use the Stefan-Boltzmann law of  
radiation:

$$P = \frac{Q}{t} = \epsilon_e A T^4$$

$$P = \left( 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4} \right) (1)$$

$$\left[ 4\pi (6.96 \times 10^8 \text{m})^2 \right] (6000\text{K})^4$$

$$P = 6.5 \times 10^{17}\text{W} \rightarrow \text{Ans}$$

Reference: GEAS 03-08 – Heat  
Transfer Mechanisms

33. D. 574

**Solution:**

Fahrenheit and Kelvin read the same  
temperature at 574.25.

Reference: GEAS 03-01 –  
Temperature and Thermal Expansion



34. A. absorption of energy as heat

**Solution:**

When liquid transforms to ice, there is no change in temperature (latent heat) and volume. Heat only leaves the system (and is absorbed by the surroundings).

Reference: GEAS 03-05 – Latent and Sensible Heat

35. C. 18 R

**Solution:**

$$\Delta T_K = \Delta T_C$$

$$\Delta T_K = 10K$$

$$\Delta T_K = \frac{5}{9} \Delta T_R$$

$$10K = \frac{5}{9} \Delta T_R$$

$$\Delta T_R = 18R \rightarrow \text{Ans}$$

Reference: GEAS 03-01 – Temperature and Thermal Expansion

36. D. melt 1 g of ice

**Solution:**

Heat of fusion is the amount of energy needed to melt ice into liquid water.

Reference: GEAS 03-05 – Latent and Sensible Heat

37. B. the energy per unit mass absorbed as heat during a phase transformation

**Solution:**

Heat of transformation is the heat of fusion. It is the energy needed for a phase change without change in temperature.

Reference: GEAS 03-05 – Latent and Sensible Heat

38. D. heat capacities

**Solution:**

$$\text{From the formula, } Q = mc(T_f - T_i)$$

if they have the same mass, the same initial temperature, and the same energy absorbed, then they must have different values of  $c$  if the final temperatures are different.

Reference: GEAS 03-05 – Latent and Sensible Heat

39. A. the temperature does not change

**Solution:**

Every time that the phase of the substance is changing when heat is applied, the temperature will never change.

Reference: GEAS 03-05 – Latent and Sensible Heat

40. D. the containing vessel also expands

**Solution:**

Since liquids are always in a container, the containing vessel also expands if you apply heat.

Reference: GEAS 03-01 – Temperature and Thermal Expansion

41. B. 360 BTU

**Solution:**

Since the car is moving AWAY from the observer, use the formula for Doppler Effect (away):

$$Q = mc\Delta T$$

$$Q = (2\text{lb}) \left( 1.0 \frac{\text{BTU}}{\text{lb}^\circ\text{F}} \right) (212^\circ\text{F} - 32^\circ\text{F})$$

$$Q = 360\text{BTU} \rightarrow \text{Ans}$$

Reference: GEAS 03-05 – Latent and Sensible Heat

42. D. A hyperbola

**Solution:**

The graph of an isothermal process is hyperbolic. For comparison, the graph of isobaric is horizontal and the graph of isochoric is vertical.

Reference: GEAS 03-07 – Thermodynamic Processes

43. D. efficiency of a Carnot engine depends only on the temperatures of the two reservoirs

**Solution:**

By the 2<sup>nd</sup> Law of Thermodynamics, the Carnot efficiency or the maximum thermal efficiency of an engine is dependent only on the temperature of the hot and cold reservoir.

Reference: GEAS 03-09 – Heat Engines

44. B. it has a lower coefficient of linear expansion

**Solution:**

If iron has a lower coefficient of expansion, its thermal expansion will be less.

Reference: GEAS 03-05 – Latent and Sensible Heat

45. A. the amount of heat energy that raises its temperature by 1° C

**Solution:**

By definition, **heat capacity (C)** is the amount of energy needed to raise the temperature of a substance by 1 degree.

This is different from **specific heat capacity (c)**. Specific heat capacity is the amount of energy needed to raise a unit mass of a substance by 1 degree. The specific heat capacity is the heat capacity divided by mass.

Reference: GEAS 03-05 – Latent and Sensible Heat

46. D. the work done by the environment on the system equals the change in internal energy

**Solution:**

According to the 1<sup>st</sup> law of Thermodynamics:  $\Delta U = \pm Q \pm W$





For an adiabatic process,  $Q = 0$ .  
That means the change in internal energy is equal to the work done on or by the system.

Reference: GEAS 03-07 –  
Thermodynamic Processes

**47. D. steel is a better heat conductor**

**Solution:**

Metals like steel, in general, have higher thermal conductivity than wood.

Reference: GEAS 03-08 – Heat  
Transfer Mechanisms

**48. B. first law of thermodynamics**

**Solution:**

An engine can never produce work and reject excess energy if there is no input energy to it because it violates the law of conservation of energy or the 1<sup>st</sup> law of thermodynamics. In other words, an engine can never create its own energy.

Reference: GEAS 03-09 – Heat  
Engines

**49. C. takes heat in, does work, and rejects heat**

**Solution:**

By definition, a heat engine is a device that takes in energy by heat and converts a fraction of that as work.

Reference: GEAS 03-09 – Heat  
Engines

**50. B. cal/(cm · s · C°)**

**Solution:**

From the Law of Conduction, the thermal conductivity  $k$  is defined as:

$$k = \frac{Q}{At \frac{\Delta T}{L}}$$

Reference: GEAS 03-08 – Heat  
Transfer Mechanisms